Multi-Armed Bandits

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Learning to Act Under Uncertainty

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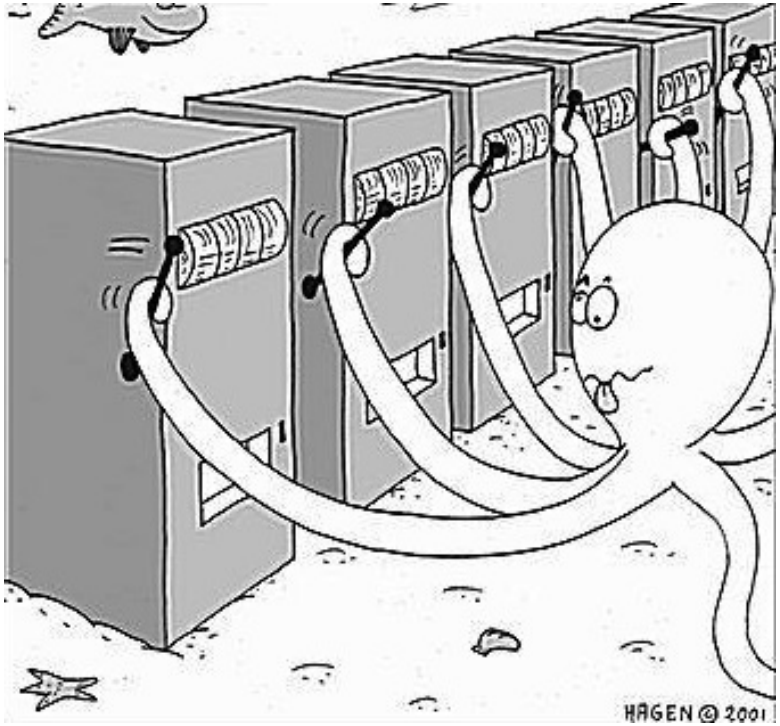
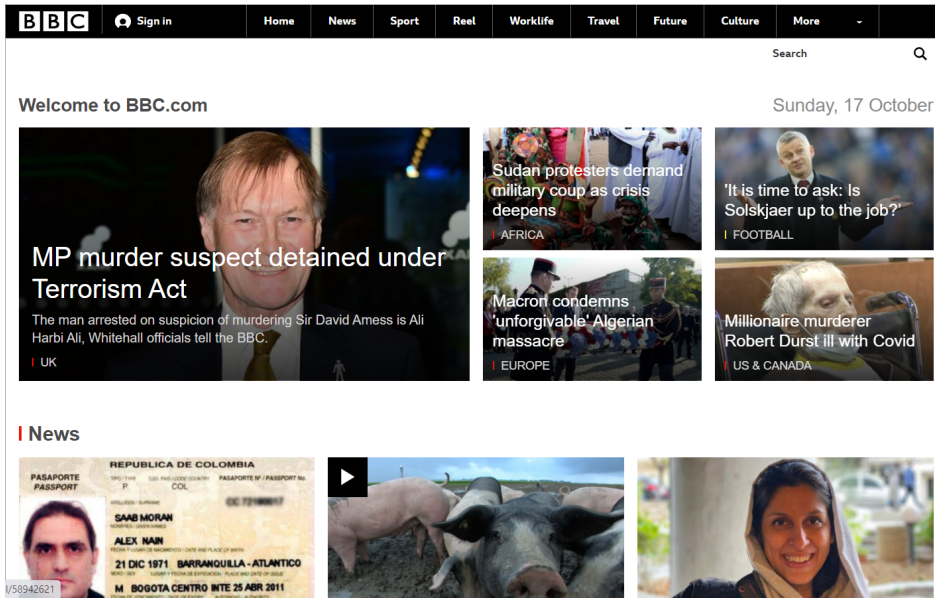


- You are standing in front of three slot machines. Each machine gives a reward with its own (unknown) probability. Some pay out more often than others.
- Your goal: figure out which arm is the best one to pull.
- A game: `https://cse442-17f.github.io/LinUCB/`

Examples

concrete albeit very stylized



News site When a new visitor arrives, the website selects article to display and observes whether the visitor clicks on it.

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Pricing When a customer enters, the store offers a price and observes whether the customer purchases or not.

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Investment Each morning, you choose one stock to invest. At the end of the day, you observe how all the stocks performed.

Canteen ?

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- K possible actions/arms
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- Each round, you choose an arm and collect a reward for this arm
- For each arm, reward is drawn **independently** from an unknown distribution which is **fixed** across time (i.i.d rewards, depends only on the chosen arm)

Back to our examples

Example	**Action**	**Reward**
News site	Choose an article to display	1 if clicked, 0 otherwise
Pricing	Offer a price p	p if purchased, 0 otherwise
Investment	Choose a stock to invest in	Change in value during the day
Canteen	?	?

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- But you learn nothing about the rewards from arms you didn't choose. So, what should we do?
- The algorithm must *explore*: try different arms to gather information.
- After all, if you always eat at the same food stall, how would you know if another one tastes better?

Exploration vs. Exploitation

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  - explore too little, and you might miss better options;
  - explore too much, and you miss out on rewards.
- A good algorithm learns which arms perform best, while limiting unnecessary exploration.

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Three Types of Feedback

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We will focus on problems with **bandit feedback**.

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- Instead of learning the best arm, we learn the best **policy**, a rule for choosing the best action **given the context**.

From Intuition to Algorithms

Reward Models

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Stochastic (non-I.I.D.)	Rewards evolve over time as a stochastic process, e.g., a random walk.

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- The best achievable mean reward is

$$\mu^* := \max_{a \in \mathcal{A}} \mu(a), \quad a^* = \arg\max_{a \in \mathcal{A}} \mu(a).$$

Problem Protocol: Multi-Armed Bandits

Setup

- K arms (actions), T rounds.
- In each round $t \in [T]$:
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- Rewards for each arm are **i.i.d.** over time.
- Per-round rewards are bounded, typically $r_t \in [0, 1]$.

How to know if an algorithm is doing a good job?

Regret

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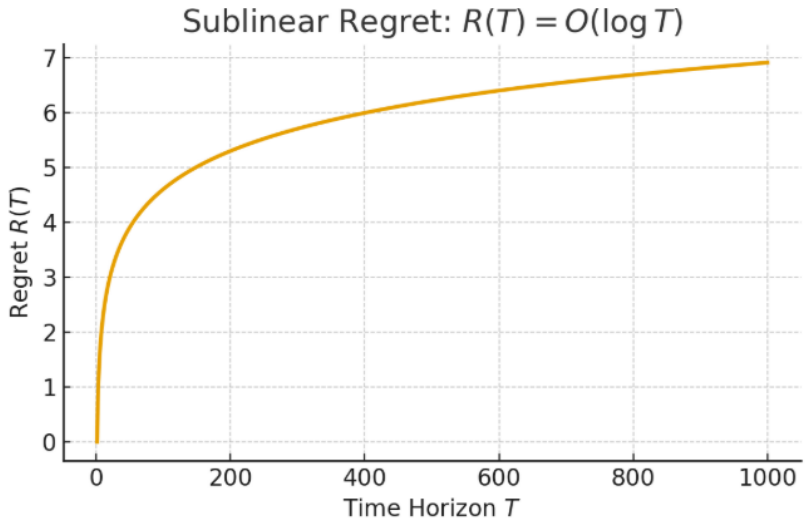

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What is the simplest algorithm you can think of?

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Try everything equally.

Uniform Exploration

Algorithm 1.1: Explore-First with parameter N

1. **Exploration phase:** try each arm N times.
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- **What could go wrong?**

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- How could we improve this algorithm?

ε-Greedy Algorithm

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- **But:** fixed ε means exploration never stops, even when we already know a good arm.
- Can we make ε *adaptive* as we learn?

Adaptive ε-Greedy

Algorithm

Use a **decaying** exploration rate ε_t:

- With probability ε_t, explore (choose an arm uniformly at random).
- With probability $1 - \varepsilon_t$, exploit (choose the best arm so far).

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Theory

For a suitable choice of $\varepsilon_t = \frac{1}{t}$, the expected regret satisfies

$$\mathbb{E}[R(T)] = O(\log T),$$

as shown in Auer et al. (2002) and Cesa-Bianchi and Fischer (1998).

Empirical Finding: Simplicity Wins

Vermorel and Mohri (2005) **found no practical advantage** in using adaptive ε-greedy variants.

> *"The most naive approach, the ε-greedy strategy, proves to be often hard to beat."*

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- Some arms are well understood, others remain uncertain.
- **Idea:** Explore when uncertain, exploit when confident.
- Instead of random exploration, explore guided by uncertainty.

Upper Confidence Bound (UCB1)

Notation

- $n_a(t)$: number of times arm a has been chosen up to round t.
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Idea

At each round, choose the arm that maximizes an **optimistic estimate** of its reward:

$$\text{UCB}_a(t) = \hat{\mu}_a(t) + \sqrt{\frac{2 \ln t}{n_a(t)}}.$$

The second term acts as an *exploration bonus* that shrinks as $n_a(t)$ grows.

Optimism in the face of uncertainty.

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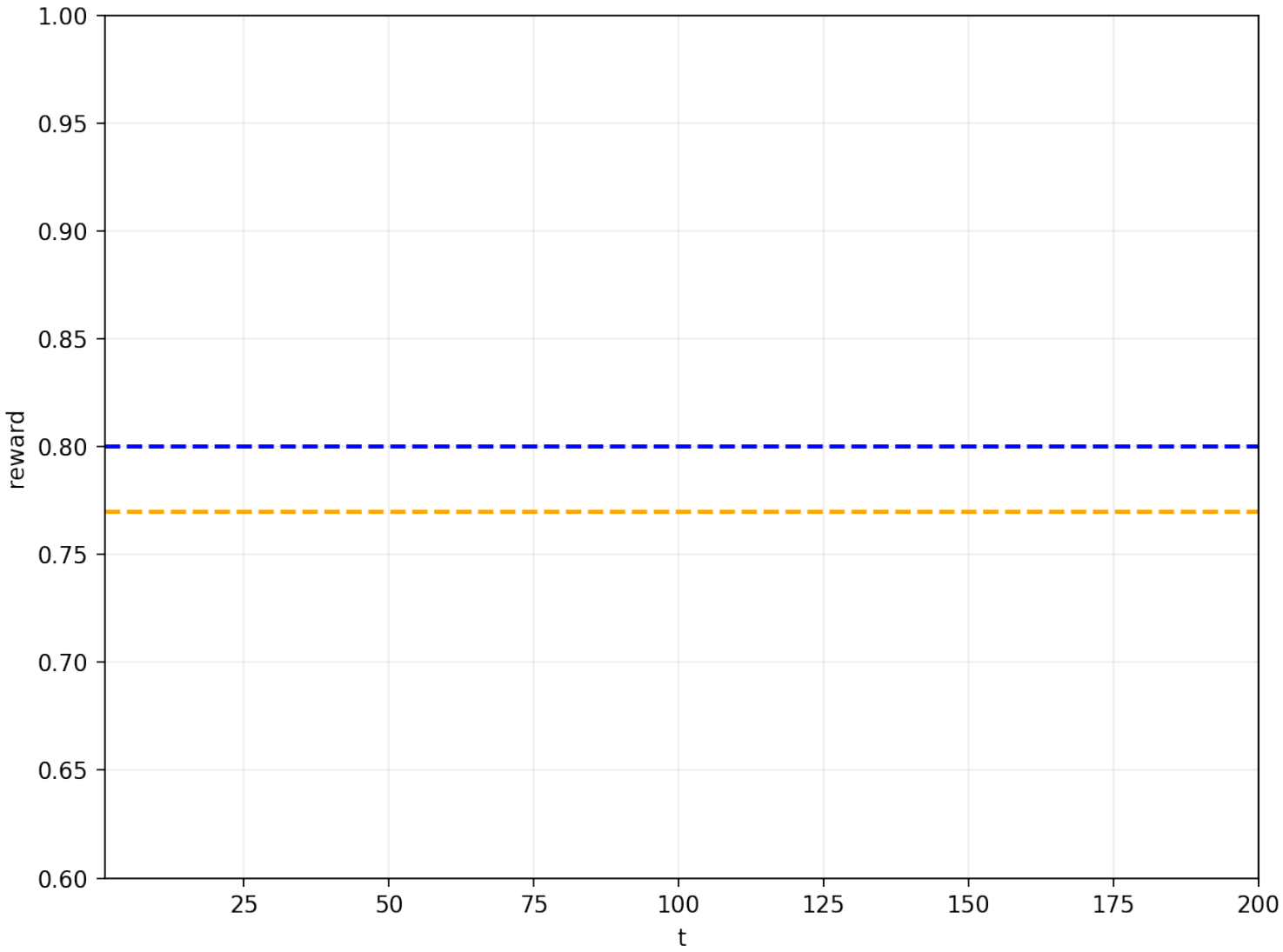

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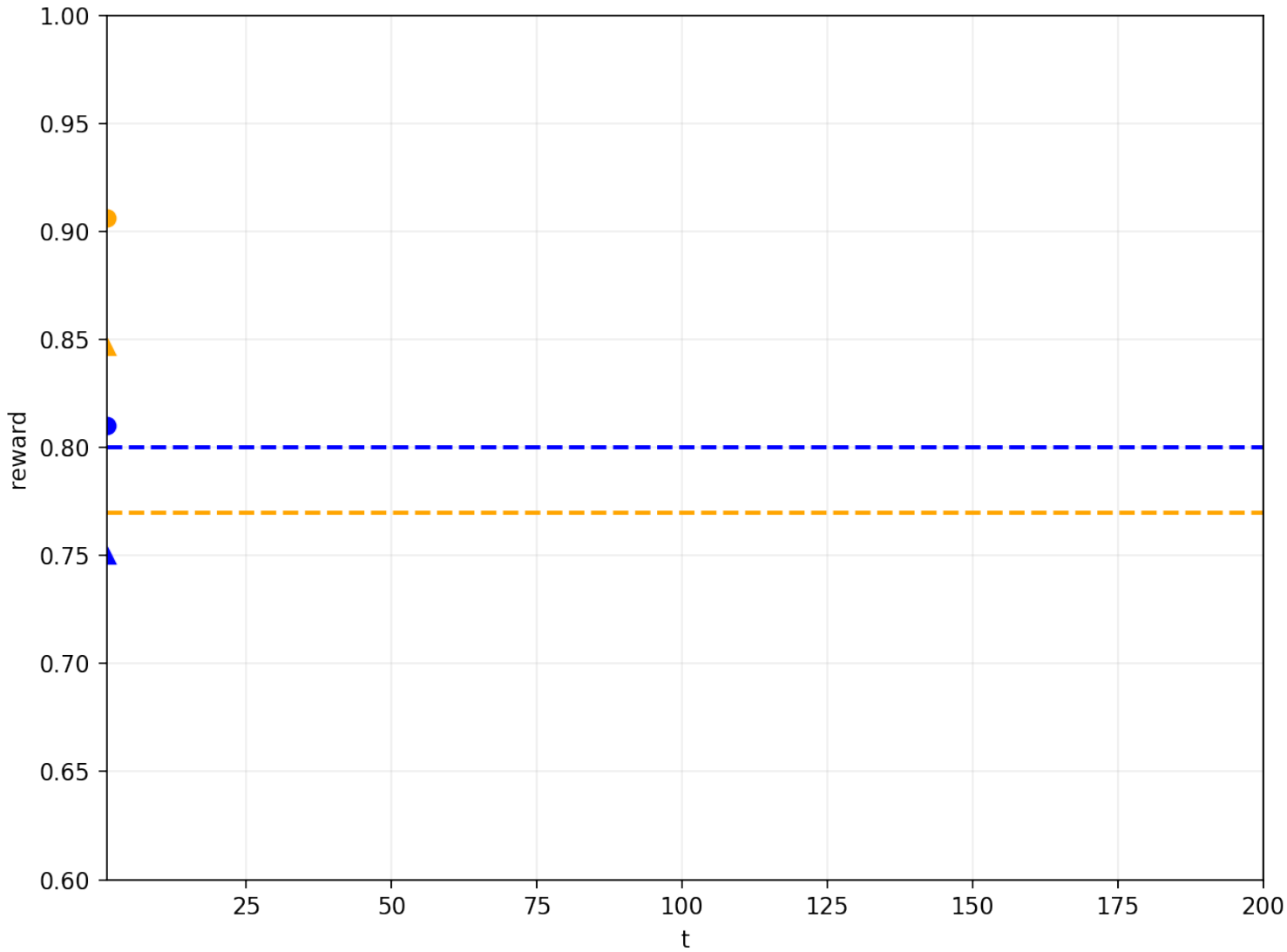

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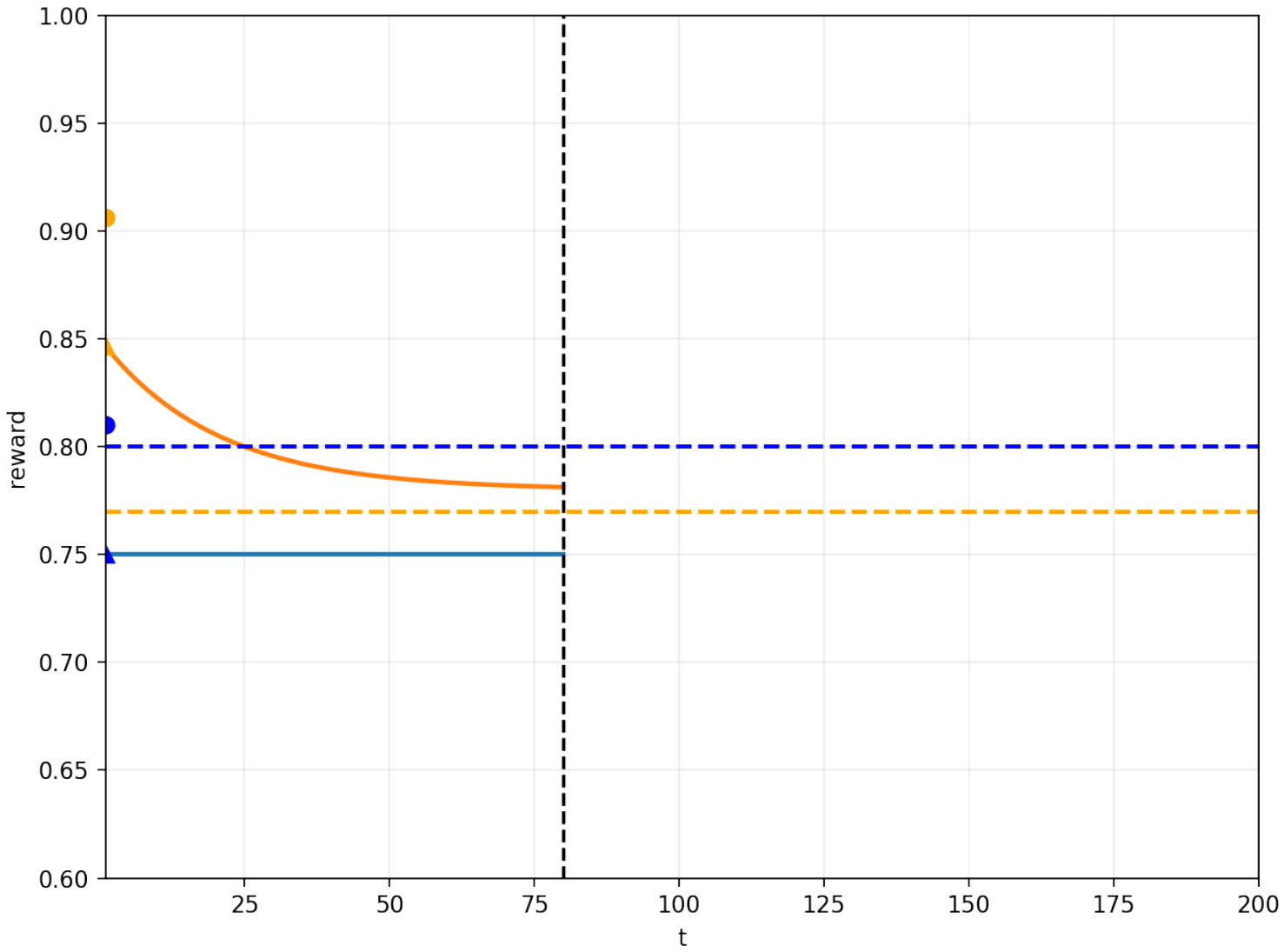

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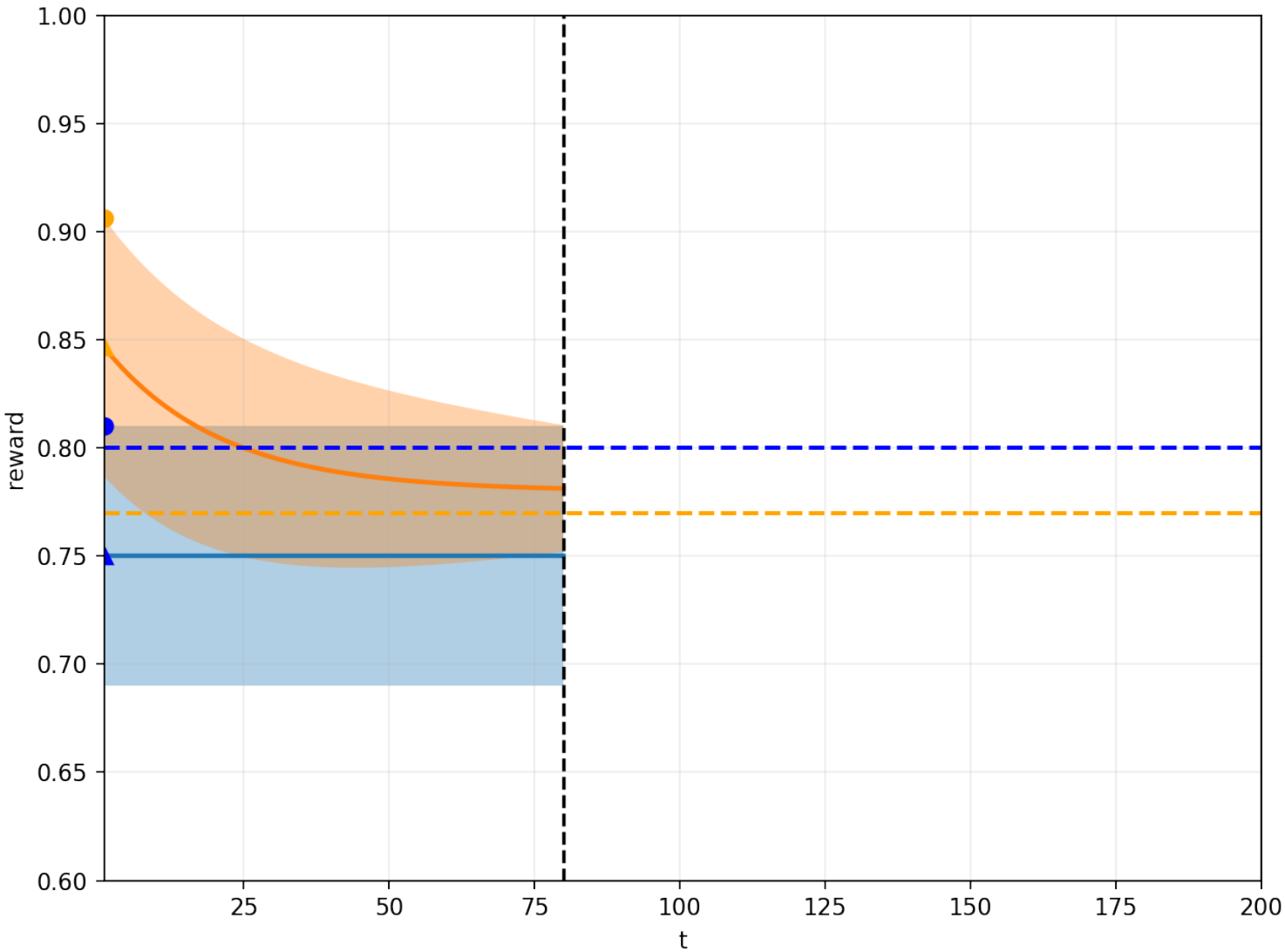

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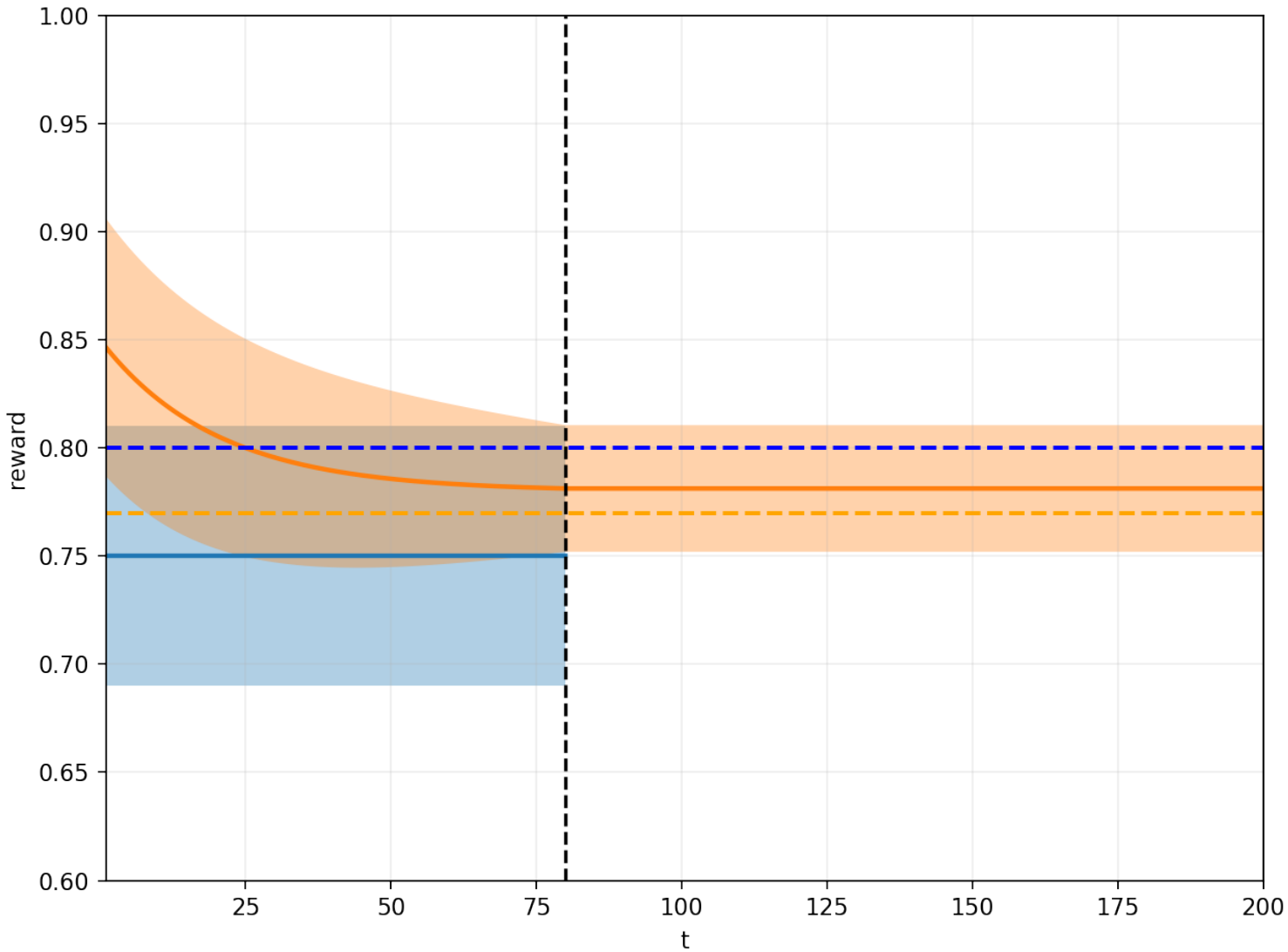

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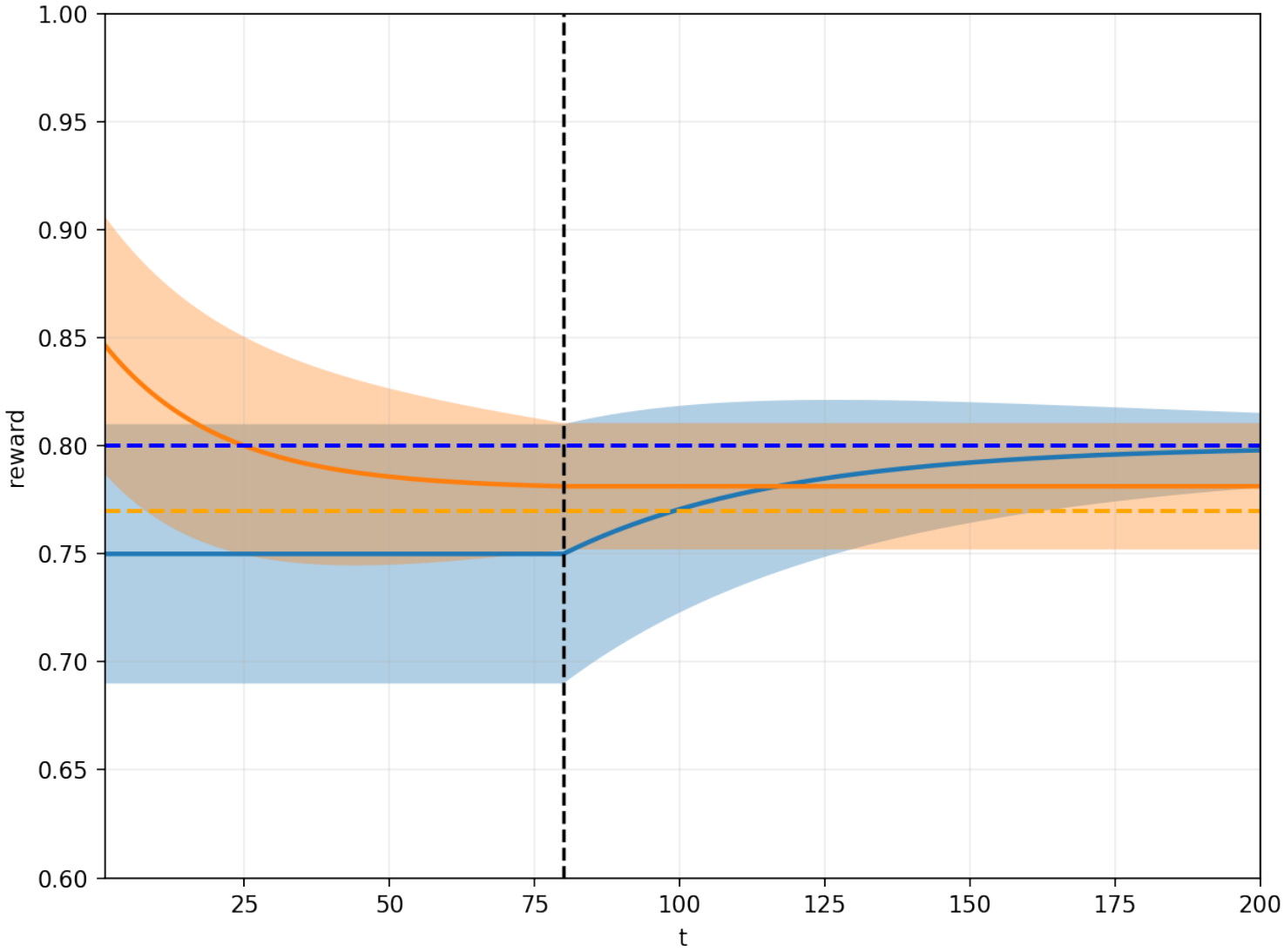

Multi-Armed Bandits in Action

Application Area	Example Bandit Problem
Healthcare	Which drug should a patient receive in a clinical trial?
E-commerce	Which price should we offer to maximize profit?
Online advertising	Which ad should we show to this user right now?
Search	How should we rank the search results for this query?
Recommender systems	Which product or movie should we recommend?
Web design	Which button color or layout gets more clicks?
Crowdsourcing	Which task should go to which worker, and at what price?
Robotics	Which strategy should the robot try in this environment?
Networking	Which settings maximize connection speed or quality?

The magic of Multi-Armed Bandits:

balancing exploration and exploitation

to make smarter, adaptive decisions.

Simple. Powerful. Universal.

References

Auer P, Cesa-Bianchi N, Fischer P (2002) Finite-time analysis of the multiarmed bandit problem. *Machine learning* 47(2-3):235–256.

Cesa-Bianchi N, Fischer P (1998) Finite-time regret bounds for the multiarmed bandit problem. *ICML*, volume 98, 100–108 (Citeseer).

Vermorel J, Mohri M (2005) Multi-armed bandit algorithms and empirical evaluation. *European conference on machine learning*, 437–448 (Springer).